

ENV S 193DS Final

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https://github.com/stellassymonds/ENVS-193DS_spring-2026_final

```
# read in packages
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(DHARMA) # for residual plots
library(ggeffects) # getting model predictions
library(MuMIn) # model selection

# read in data
nests <- read.csv(here("data", "nest_data_final.csv"))
```

Problem 1

- a. In part 1, they used a generalized linear model. In part 2, they used a chi-squared test.
- b. Part 1: We found evidence to suggest that the distance to water edge does predict probability of American Avocet microhabitat use.

Part 2: We found evidence to suggest that there is a difference between American Avocet, Forster's Tern, and Black-necked Stilt bird species' habitat preferences of managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat habitats.

- c. Associated behavior type of either feeding, breeding, or roosting could be a categorical variable that influences the probability of American Avocet microhabitat use. For example, waterbirds that are breeding or roosting may prefer islands within ponds.

Problem 2

a.

```
# create new object from nests
nests_clean <- nests |>
  # reordering ant species
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB", # C. sjostedti
                                   "RRB", # C. mimosae
                                   "BBR")) |> # C. nigriceps

# select for preferred columns
select(height_cm, ant_species, case_control) |>
# filter for preferred ant species
filter(ant_species %in% c("RRB", "BBR", "AB")) |>
# create new column with full scientific name
mutate(scientific_name = case_when(
  ant_species == "AB" ~ "C. sjostedti",
  ant_species == "RRB" ~ "C. mimosae",
  ant_species == "BBR" ~ "C. nigriceps"))

slice_sample(nests_clean, # data frame to display
             n = 10) # select ten random rows
```

	height_cm	ant_species	case_control	scientific_name
1	97.5	RRB	0	C. mimosae
2	157.5	BBR	1	C. nigriceps
3	71.0	BBR	0	C. nigriceps
4	181.5	BBR	1	C. nigriceps
5	168.0	RRB	0	C. mimosae
6	201.0	BBR	1	C. nigriceps
7	236.5	RRB	0	C. mimosae
8	391.5	RRB	1	C. mimosae
9	156.0	AB	0	C. sjostedti
10	223.5	RRB	0	C. mimosae

- b. 1s mean that the tree contained a nest and 0s mean the tree did not contain a nest and was available for comparison.
- c. Tree height is a continuous variable. As tree height increases, the probability of nest occurrence would increase because they are farther from possible nest predators or disturbances (Results, paragraph 2).

Acacia ant species is a categorical variable. If an ant species is a more aggressive defender, the probability of nest occurrence would increase because the ant species reduces the risk of nest predation (Introduction, paragraph 3). If acacia ants defend against all possible tree disturbances, then the probability of nest occurrence in trees with less aggressive ant species would increase (Introduction, paragraph 3).

- d. 2 models total:

Model number	Tree height	Ant Species	Interaction?
1	X	X	no
2	X	X	yes

- e.
- f.

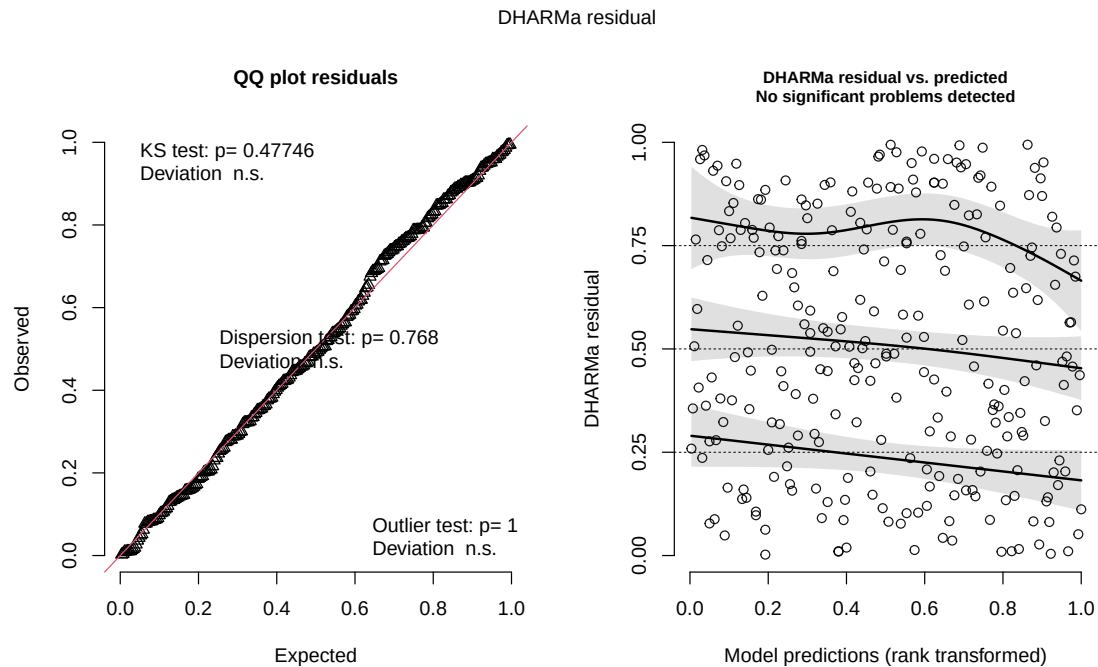
```
AICc( # calculating AIC of both models
  model1,
  model2) |>
# arranging output in order of AIC
arrange(AICc)
```

```
      df    AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) shows nest occurrence as a function of tree height in cm interacting with ant species.

- g.

```
# generating diagnostic plots for best model (by AIC value) from DHARMA
plot(
  simulateResiduals(model2)
)
```



h.

```

preds <- ggpredict(
  model2, # specify model
  terms = c("height_cm", "ant_species")) |>
  # renaming columns
  rename(case_control = x,
          ant_species = group)

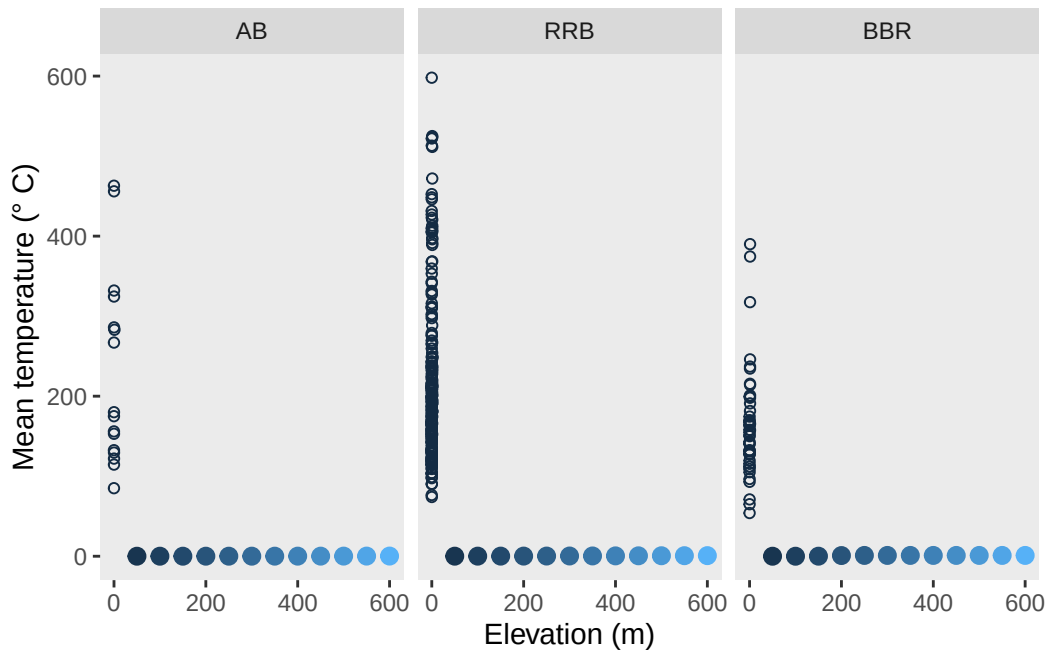
# creating visualization
ggplot(data = nests_clean,
       mapping = aes(x = case_control,
                     y = height_cm,
                     color = case_control)) +
  # create separate panels for each species
  facet_wrap(~ ant_species) +
  geom_jitter(height = 0,
             width = 0.1,
             shape = 21) +
  geom_pointrange(data = preds,
                 mapping = aes(x = case_control,
                               y = predicted,

```

```

      ymin = conf.low,
      ymax = conf.high)) +
theme(legend.position = "none") + # take out legend
labs(x = "Elevation (m)",
     y = "Mean temperature (\U00B0 C)") +
# take out gridlines and gray background
theme(panel.grid = element_blank())

```



i. Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>

j.

k.

Problem 3

a. My affective visualization would be impossible to analyze without a key accompanying it. I prioritized creativity over feasibility of understanding. My exploratory visualizations are very easy to understand and make conclusions about. All my visualizations showed close

to an equal amount of bike rides when I did and didn't have plans after class and low, medium, or high wind speed. The variation in data was consistent and can be seen in all visualizations. A slight difference in medians between bike rides when I have plans after class and no plans after class is visible in my exploratory visualization, but is not clear in my affective visualization. This is because my data is oriented around a central point in my affective visualization. I don't think a clear association between wind speed and bike ride duration is obvious in any of my visualizations. My feedback in workshop consisted of possibly adding icons to the end of the lines to represent temperature or another variable. I ultimately decided to not implement this feedback, since neither temperature nor other variables I tracked had any visible affect on my data, and I thought it may make it harder to see the difference between the lengths of each stitched line.

- b. This analysis is appropriate to find a correlation between status of my plans after class and my bike ride durations. one variable is binary
- c.